Release check and independent reproduction

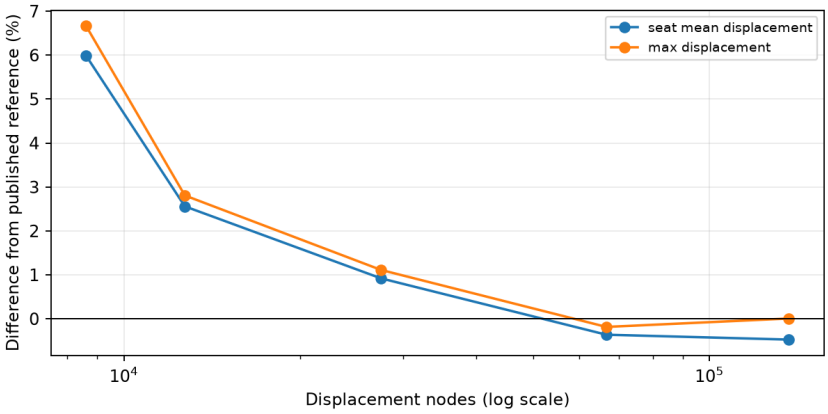
Baseline verified: all 588 files listed in the v1.1 checksum manifest match. The frozen release still contains 118 active tasks and 298 published targets. This supplement adds independent validation evidence and 14 unscored source pairs.

Two displacement tasks reproduced: independent Gmsh remeshing and FEniCSx quadratic-displacement solves on 11 meshes. Reference values were read only by the comparison stage; none were replaced by new solver outputs.

A-AGENTFEM-fresh_test_000

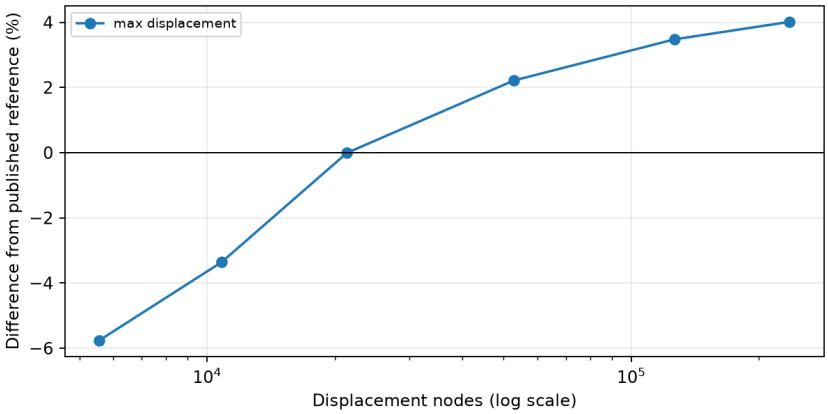
seat_mean_displacement_mm: published 0.098354836 mm; independent 0.097888916 mm; difference 0.474%; last refinement change 0.111%.

max_displacement_mm: published 0.17562809 mm; independent 0.17563918 mm; difference 0.006%; last refinement change 0.192%.



A-CYC-Coupler-PETGCF

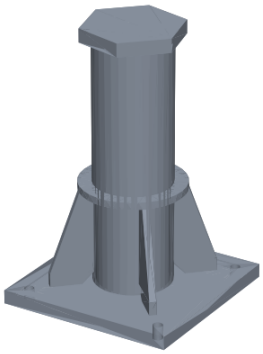
max_displacement_mm: published 0.0076174664 mm; independent 0.0079230724 mm; difference 4.012%; last refinement change 0.515%.

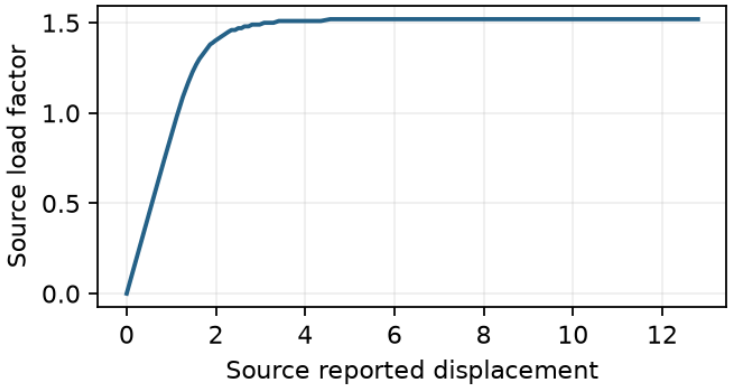


Limits: requested displacements only; no validation of every stress field or source family. Last-mesh change is an empirical indicator, not a rigorous error bound. The other 116 active tasks still lack independent reproduction.

New source CAD and published nonlinear curves

Wyk Rohrstoss ring

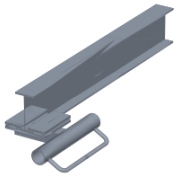


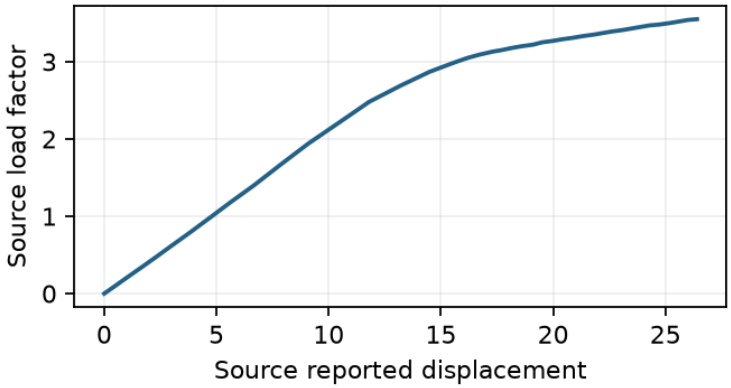


Original FreeCAD model: 12 solid regions, 89 faces. Published analysis: elastic-plastic, geometric linear; 61 printed history rows. Solid regions may be boolean fragments, not separate assembly parts.

CAD view is a read-only tessellation of the original stored BREP. Curve is plotted directly from the original .out table; no new nonlinear FEM solution was generated.

Haltegriff FEM RC





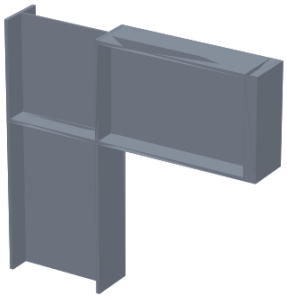
Original FreeCAD model: 11 solid regions, 91 faces. Published analysis: elastic-plastic, geometric non-linear; 41 printed history rows. Solid regions may be boolean fragments, not separate assembly parts.

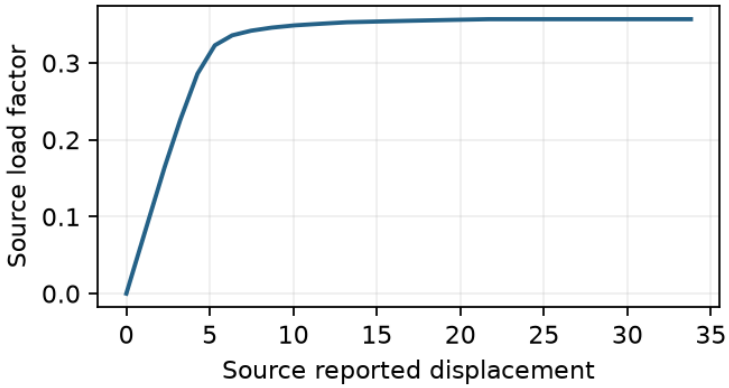
CAD view is a read-only tessellation of the original stored BREP. Curve is plotted directly from the original .out table; no new nonlinear FEM solution was generated.

Source: github.com/HarryvL/fcVM-workbench. Raw files, pinned Git blob hashes, native properties and exact table copies are included. Source-pair status: held for setup/revision mapping and numerical validation.

Additional complex structural candidates

IPE400 4

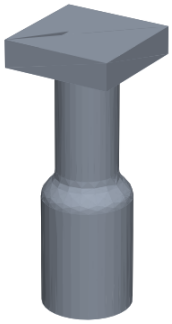


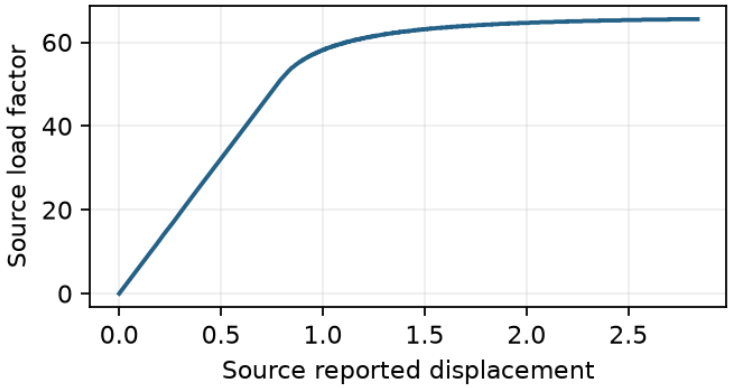


Original FreeCAD model: 7 solid regions, 68 faces. Published analysis: elastic-plastic, geometric non-linear; 21 printed history rows. Initial imperfections and control-file material overrides must be matched to this specific output.

The load factor scales a source-defined loading pattern; it is not a force in newtons. Printed stresses are associated with the source-selected critical Gauss point, not automatically global stress maxima.

transition





Original FreeCAD model: 4 solid regions, 17 faces. Published analysis: elastic-plastic, geometric linear; 501 printed history rows. Initial imperfections and control-file material overrides must be matched to this specific output.

The load factor scales a source-defined loading pattern; it is not a force in newtons. Printed stresses are associated with the source-selected critical Gauss point, not automatically global stress maxima.

Published CAD and curves are preserved unchanged. These figures are inspection views, not dimensioned manufacturing drawings or newly generated benchmark problems.

Inventory, admission limits and remaining work

14 native CAD/result pairs: 14 valid extracted analysis BREP shapes; 996 published table rows; five original VTK result files. Named cases are not counted as independent physical families. Beam variants and transition revisions remain grouped for breadth claims.

- **Bar buckling** - 11 history rows; original VTK available
- **Buckling** - 1 history rows; original VTK available
- **buckling_U100_3D** - 22 history rows; original VTK available
- **tensile** - 18 history rows; original VTK available
- **Lug_Pressure_Example** - 61 history rows; original VTK available
- **barrier** - 201 history rows; original printed result table
- **box girder** - 21 history rows; original printed result table
- **train_bridge** - 15 history rows; original printed result table
- **transition** - 501 history rows; original printed result table
- **transition3** - 1 history rows; original printed result table
- **Plate_girder_EC3** - 21 history rows; original printed result table
- **Wyk_Rohrstoss_ring** - 61 history rows; original printed result table
- **Haltegriff_FEM_RC** - 41 history rows; original printed result table
- **IPE400_4** - 21 history rows; original printed result table

Admission remains pending. Before scoring these nonlinear cases, match the native analysis and controls to each output, establish the displacement/imperfection convention, verify load and material overrides, and set justified tolerances. A source label or valid CAD file alone is insufficient.

Code_Aster review: the existing 50-case pool provides 761 literal result-check blocks and 28 geometry scripts, but zero native STEP files. It remains a separate source pool; no scripts were executed and no replacement CAD was generated.

What remains: 116 active cases need independent reproduction; these 14 source pairs need admission work; contact, crash, forming and fatigue coverage is still absent from the active scored set. Existing tolerance values remain provisional, with new empirical support only for the two tested displacement tasks.

All primary references remain source-published. The new FEniCSx fields are clearly labeled validation-only. Candidate-visible benchmark inputs and hidden published answers in the frozen v1.1 release were not modified.